

# R Basic Course

Mardones, M. & Zúñiga, MJ.

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# About Course\_\_R

Description and educational material for introductory course on R, descriptive statistics, and visualizations with applications in fisheries.

This course has these sections;

- 1. **useR**. Intro to R and basic features,
- 2. Exploratory Data Analysis, Data handling, Data manipulation
- 3. Descriptive Statistics. Parametric and No Parametric, Linear Regression, among others
- 4. Visualization. Graphics and Maps using modern packages ... and more

This is a *sample* book written in **Markdown**. You can use anything that Pandoc's Markdown supports; for example, a math equation  $a^2 + b^2 = c^2$ .

## Usage

Each **bookdown** chapter is an .Rmd file, and each .Rmd file can contain one (and only one) chapter. A chapter *must* start with a first-level heading: **# A good chapter**, and can contain one (and only one) first-level heading.

Use second-level and higher headings within chapters like: **## A short section** or **### An even shorter section**.

The **index.Rmd** file is required, and is also your first book chapter. It will be the homepage when you render the book.

## Render book

You can render the HTML version of this example book without changing anything:

1. Find the **Build** pane in the RStudio IDE, and

2. Click on **Build Book**, then select your output format, or select “All formats” if you’d like to use multiple formats from the same book source files.

Or build the book from the R console:

To render this example to PDF as a `bookdown::pdf_book`, you’ll need to install XeLaTeX. You are recommended to install TinyTeX (which includes XeLaTeX): <https://yihui.org/tinytex/>.

## Preview book

As you work, you may start a local server to live preview this HTML book. This preview will update as you edit the book when you save individual `.Rmd` files. You can start the server in a work session by using the RStudio add-in “Preview book”, or from the R console:

# Chapter 1

## useR. Intro to R and basic features

Source in this link

An introduction to R





## Chapter 2

# Exploratory Data Analysis, Data handling, Data manipulation

Cross-references make it easier for your readers to find and link to elements in your book.

### 2.1 Chapters and sub-chapters

There are two steps to cross-reference any heading:

1. Label the heading: `# Hello world {#nice-label}`.
  - Leave the label off if you like the automated heading generated based on your heading title: for example, `# Hello world = # Hello world {#hello-world}`.
  - To label an un-numbered heading, use: `# Hello world {-#nice-label}` or `{# Hello world .unnumbered}`.

### 2.2 Captioned figures and tables

Figures and tables *with captions* can also be cross-referenced from elsewhere in your book using `\@ref(fig:chunk-label)` and `\@ref(tab:chunk-label)`, respectively.

See Figure 2.1.

Don't miss Table 2.1.

Sources

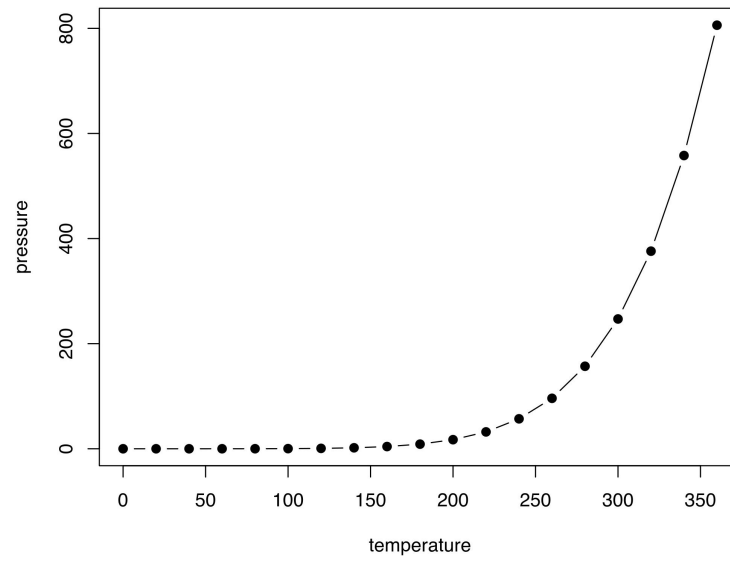


Figure 2.1: Here is a nice figure!

Table 2.1: Here is a nice table!

| temperature | pressure |
|-------------|----------|
| 0           | 0.0002   |
| 20          | 0.0012   |
| 40          | 0.0060   |
| 60          | 0.0300   |
| 80          | 0.0900   |
| 100         | 0.2700   |
| 120         | 0.7500   |
| 140         | 1.8500   |
| 160         | 4.2000   |
| 180         | 8.8000   |

Basic R

Data Explorer

EMSE 4572 CleanData